MLOPS ASSIGNMENT -02

NAME :MANASA G R

USN:4GW23CI027

1. Which run had the lowest latency, and why might that be?

run_batch8 recorded the lowest latency at 0.276 s (0.27556). run_batch1 recorded a higher latency of 0.878 s (0.87846). This represents an approximate 68.6% drop in execution latency when switching from batch size 1 to batch size 8.

2. Did accuracy change between runs? Why or why not?

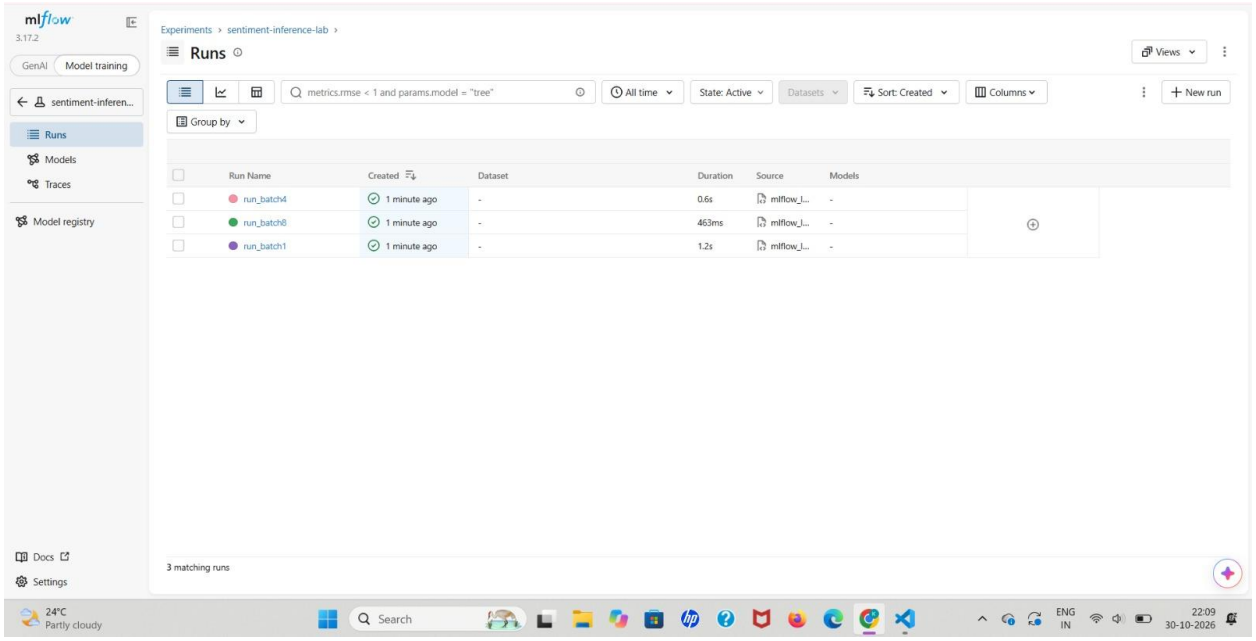
No, accuracy did not change. It remained constant at 1.0 (100%) across both runs.

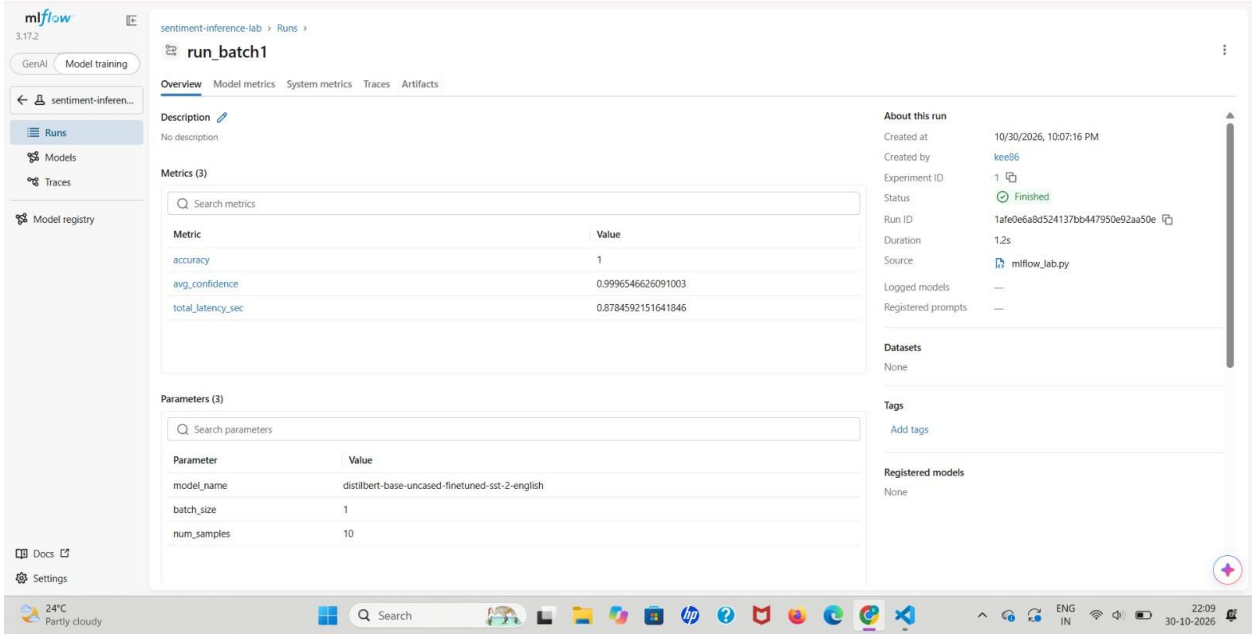
Deterministic inference: Both runs evaluated the exact same pre-trained model (distilbert-base-uncased-finetuned-sst-2-english) using identical frozen weights on the same set of 10 evaluation sentences. Role of batch size: Batch size only affects how samples are grouped together during the forward pass to optimize throughput and execution speed; it does not alter the underlying model parameters, weights, or prediction logic.

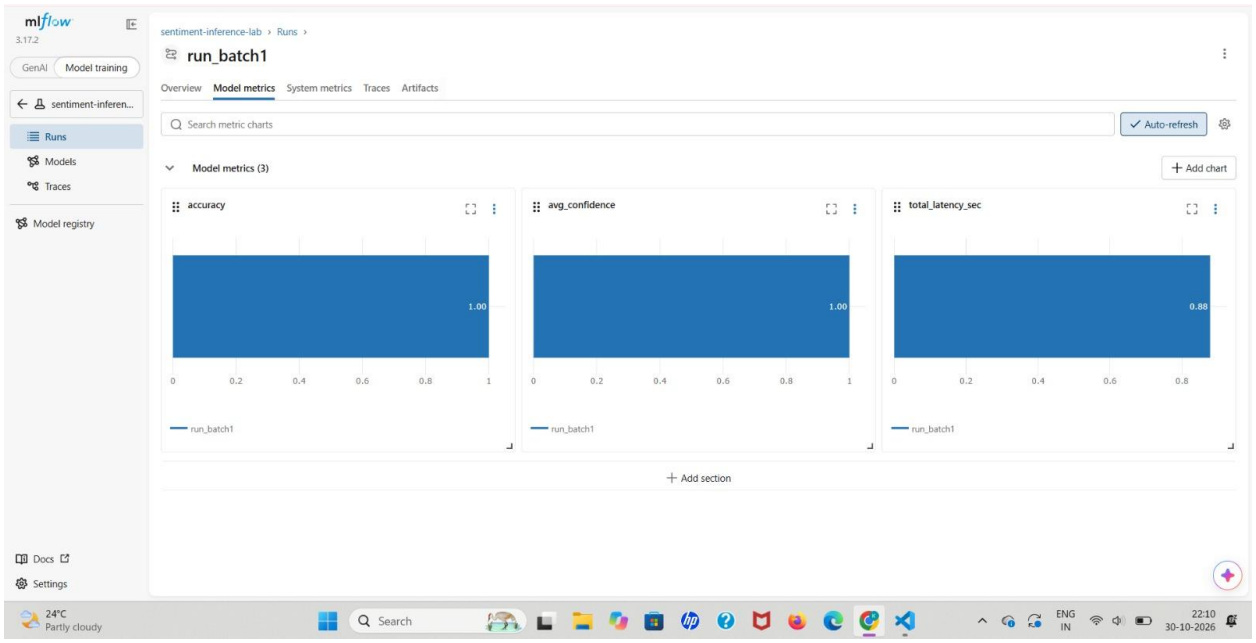
3. Name two benefits of using MLflow here instead of printing results in the terminal.

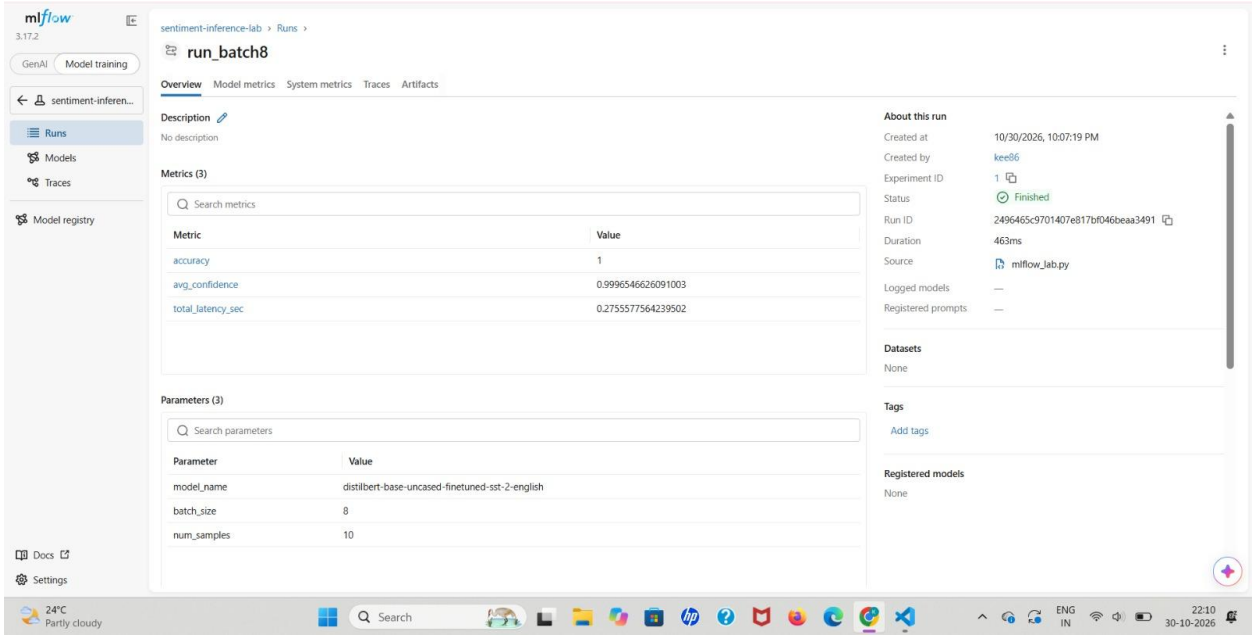
Persistent tracking and artifact storage: MLflow automatically logs hyperparameters (batch_size, model_name), performance metrics (accuracy, total_latency_sec), and output artifacts (such as run_batch1_predictions.csv and run_batch8_predictions.csv) directly to a database/file store. Terminal outputs are ephemeral and lost as soon as the session closes or the window clears.

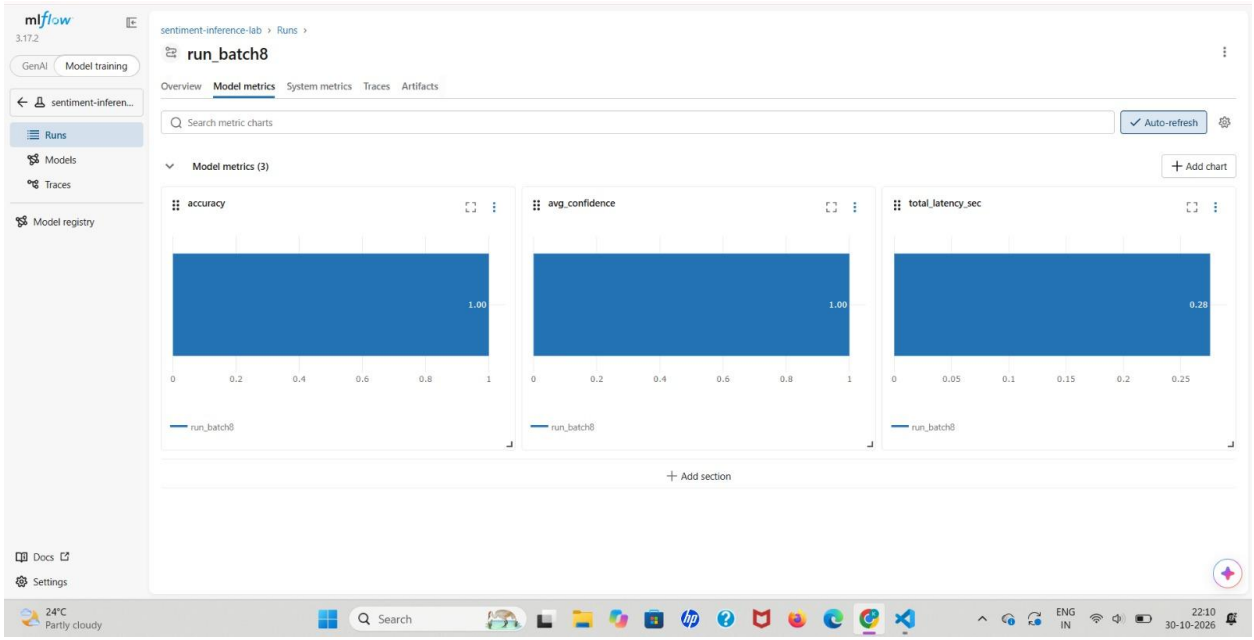
Sreenshot of the MLflow runs table:



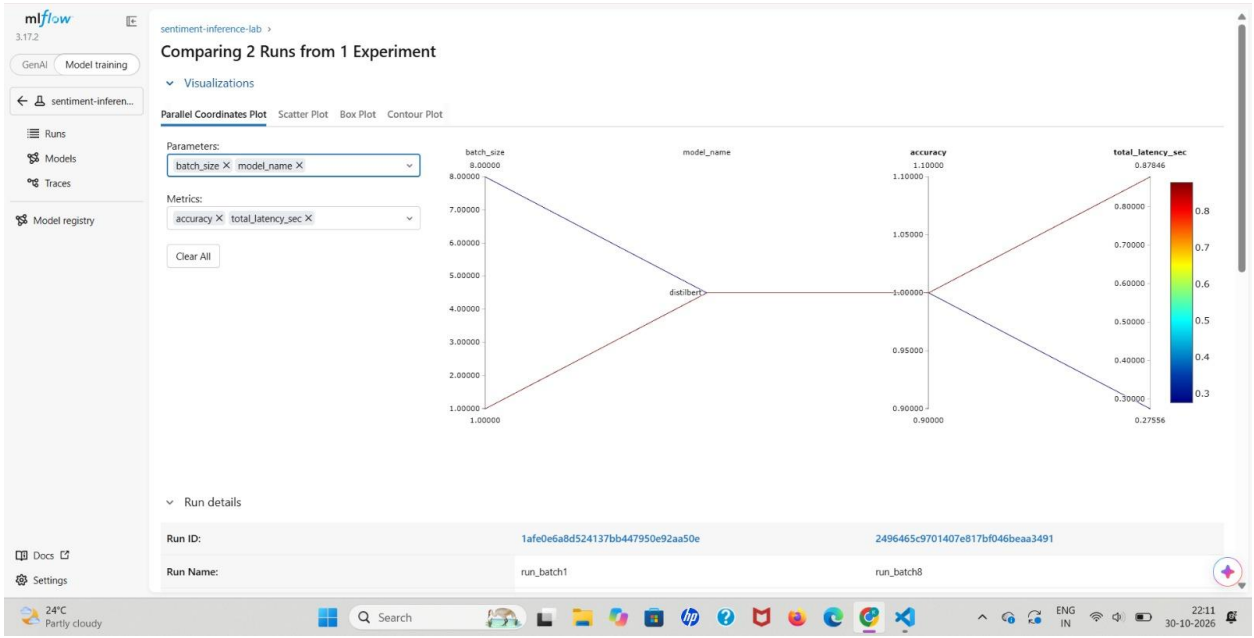



mlflow
3.17.2
GenAI
Model training
sentiment-inferen...
Runs
Models
Traces
Model registry
Docs
Settings
sentiment-inference-lab › Runs ›
run_batch1
Overview
Model metrics
System metrics
Traces
Artifacts
Search metric charts
Auto-refresh
Model metrics (3)
Add chart
accuracy
avg_confidence
total_latency_sec
1.00
1.00
0.88
0
0.2
0.4
0.6
0.8
1
run_batch1
Add section
24°C
Partly cloudy
Search
ENG
IN
22:10
30-10-2026


mlflow
3.17.2
GenAI
Model training
sentiment-inferen...
Runs
Models
Traces
Model registry
Docs
Settings
sentiment-inference-lab › Runs ›
run_batch8
Overview
Model metrics
System metrics
Traces
Artifacts
Description
No description
Metrics (3)
Search metrics
Metric
Value
accuracy
1
avg_confidence
0.9996546626091003
total_latency_sec
0.2755577564239502
Parameters (3)
Search parameters
Parameter
Value
model_name
distilbert-base-uncased-finetuned-sst-2-english
batch_size
8
num_samples
10
About this run
Created at
10/30/2026, 10:07:19 PM
Created by
kee86
Experiment ID
1
Status
Finished
Run ID
2496465c9701407e817bf046beaa3491
Duration
463ms
Source
mlflow_lab.py
Logged models
—
Registered prompts
—
Datasets
None
Tags
Add tags
Registered models
None
24°C
Partly cloudy
Search
ENG
IN
22:10
30-10-2026



Screenshot of the Compare View:



Mlflow_lab.py file:

